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MineShield

Smart Safety Monitoring System

User Manual

Version 2.0

May 20, 2026

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1 Project Overview

MineShield is a smart safety monitoring system designed to improve workplace safety through real-time hazard reporting, sensor monitoring, and alert management. The system ensures faster response times and enhanced situational awareness across all operational roles.

1.1 Core Features

- **Hazard Reporting** Immediate capture and submission of unsafe conditions with location and photo evidence.
- **Sensor Monitoring** Real-time data from IoT sensors (gas, air quality, vibration, noise).
- **Alerts and Dashboard Systems** Centralized dashboard with active alerts, zone maps, and analytics.

1.2 Technologies Used

- **Frontend:** React Native cross-platform mobile framework.
- **Backend:** Firebase authentication, Firestore database, cloud messaging, and storage.

2 Team Structure

Role	Assigned
Project Manager	Simon
Lead Developer	Klim
Firebase Lead	Frieda
UI/UX Lead	Tegameno
Documentation Lead	Andre
GitHub Manager	Rauna

3 Role Breakdown

3.1 Project Manager

- Coordination of all roles
- Timeline management
- Ensuring deliverables alignment
- Risk and issue handling

3.2 Lead Developer

- Oversees system architecture
- Ensures feature implementation (FR-001 to FR-015)
- Integrates frontend and backend
- Maintains code quality

3.3 Firebase Lead

- Designs Firestore database: Users, Hazards, Alerts, Zones
- Handles authentication and real-time database
- Implements cloud messaging
- Ensures data integrity

3.4 UI/UX Lead

Designs intuitive, accessible, and user-friendly interfaces for workers and supervisors.

3.5 Documentation Lead

Responsible for SRS documentation, consistency, and requirement traceability.

3.6 GitHub Manager

Manages repository, version control, and team collaboration.

4 System Architecture Diagram

The architecture follows a client-server model with React Native frontend and Firebase backend. (Refer to the separate architecture diagram document provided by the development team.)

5 Complete Folder Structure

The MineShield React Native project follows a modular structure for maintainability and scalability.

5.1 Source Directory (src/)

```
src/
|-- screens/
|   |-- auth/
|   |-- worker/
|   |-- supervisor/
|   |-- office/
|   |-- visitor/
|-- components/
|   |-- common/
|       |-- hazards/
|       |-- maps/
|       |-- alerts/
|       |-- charts/
|-- navigation/
|-- services/
|-- contexts/
|-- hooks/
```

5.1.1 Detailed Folder Tree

```
src/
|-- screens/
|   |-- auth/
|       |-- LoginScreen.js
|       |-- RegisterScreen.js
|       |-- ForgotPasswordScreen.js
|   |-- worker/
|       |-- WorkerDashboardScreen.js
|       |-- HazardFormScreen.js
```

```
| | |-- HazardDetailScreen.js
| | |-- AlertsScreen.js
| | |-- SensorsScreen.js
| | |-- ZoneMapScreen.js
| | |-- ProfileScreen.js
| | |-- NotificationScreen.js
| | |-- SettingsScreen.js
| | |-- AboutScreen.js
| | |-- HelpScreen.js
| |-- supervisor/
| | |-- SupervisorDashboardScreen.js
| | |-- AssignTasksScreen.js
| | |-- ReportsScreen.js
| |-- office/
| | |-- OfficeDashboardScreen.js
| | |-- AnalyticsScreen.js
| | |-- UserManagementScreen.js
| |-- visitor/
| | |-- VisitorDashboardScreen.js
| | |-- VisitorHazardsScreen.js
| | |-- VisitorAlertScreen.js
| |
|-- components/
| |-- common/
| | |-- Button.js
| | |-- InputField.js
| | |-- Card.js
| | |-- LoadingSpinner.js
| | |-- Header.js
| | |-- BottomTabBar.js
| | |-- SafeAreaView.js
| |-- hazards/
| | |-- HazardCard.js
| | |-- HazardFilter.js
| | |-- HazardStatusBadge.js
| | |-- ImageUploader.js
| |-- maps/
| | |-- HazardMarker.js
| | |-- ZoneOverlay.js
| | |-- MapControls.js
| |-- alerts/
| | |-- AlertBanner.js
| | |-- SOSModal.js
| | |-- NotificationCard.js
| |-- charts/
| | |-- BarChart.js
| | |-- LineChart.js
| | |-- PieChart.js
| |
|-- navigation/
| |-- AppNavigator.js
```

```
| |-- AuthNavigator.js
| |-- WorkerNavigator.js
| |-- SupervisorNavigator.js
| |-- OfficeNavigator.js
| |-- VisitorNavigator.js
| |-- RootNavigator.js
| |
|-- services/
| |-- firebase.js
| |-- authService.js
| |-- firestoreService.js
| |-- storageService.js
| |-- notificationService.js
| |-- sensorService.js
| |-- locationService.js
| |
|-- contexts/
| |-- AuthContext.js
| |-- HazardContext.js
| |-- AlertContext.js
| |-- ThemeContext.js
| |
|-- hooks/
| |-- useAuth.js
| |-- useHazards.js
| |-- useAlerts.js
| |-- useSensors.js
| |-- useLocation.js
```

5.2 Root Directory Files

App.js
package.json
README.md
firebase.json

A Document Revision History

Version	Date	Changes
1.0	2025-01-15	Initial SRS release
2.0	May 20, 2026	Updated folder structure and role definitions

B Contact Information

For technical support or further documentation, please contact the Documentation Lead (Andre) or the Project Manager (Simon).